NEXT.JS: FEATURES & PRACTICAL DEMONSTRATION

Salford & Co.

WHAT IS NEXT.JS?

- A React framework for production
- Supports SSR, SSG, ISR
- Built-in API routes and Image Optimization
- SEO-friendly and fast performance

KEY FEATURES OF NEXT.JS

- Server-Side Rendering (SSR)
- Static Site Generation (SSG)
- Incremental Static Regeneration (ISR)
- API Routes
- Automatic Image Optimization
- File-based Routing
- Middleware Support

ADVANTAGES OF NEXT.JS

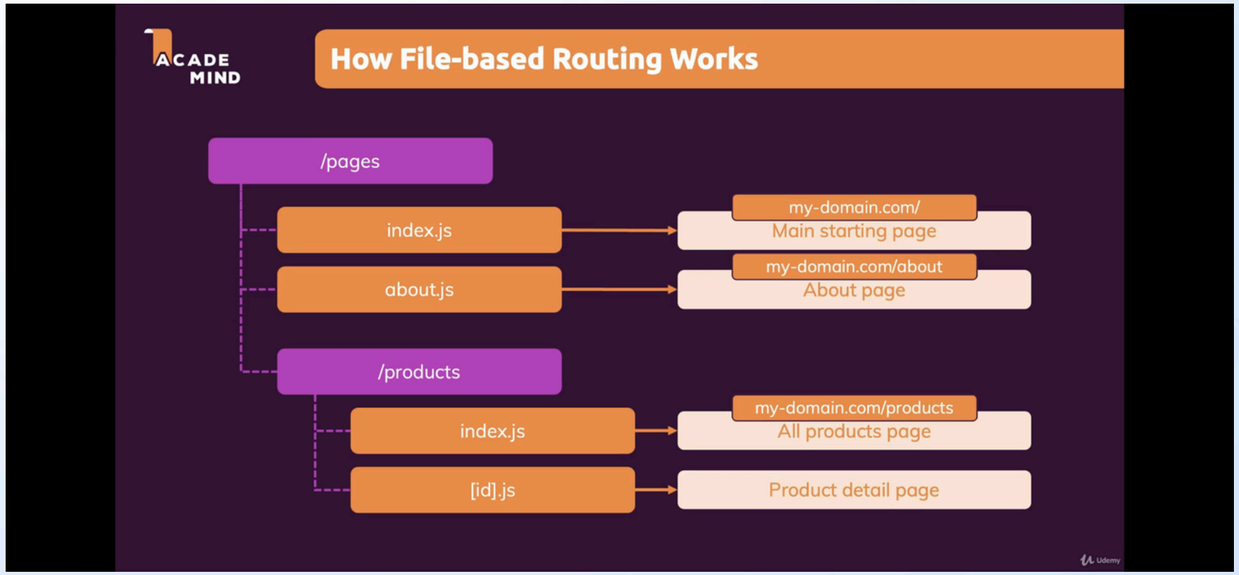
- Improved performance with SSR & SSG
- Built-in support for API routes
- Optimized images and static assets
- SEO-friendly with meta tag management
- Easy deployment with Vercel

INSTALLATION & SETUP

- Install Node.js
- Run: **npx create-next-app@latest my-app**
- Navigate: **cd my-app**
- Start: **npm run dev**
- Open in browser: **http://localhost:3000**

UNDERSTANDING FILE-BASED ROUTING

- Pages are created inside **/pages/**
- Each file represents a route
- Nested routes via folder structure
- Example: **pages/products/index.js → /products**
- Dynamic routes: **[id].js** for **/products/123**



DEMO: DYNAMIC ROUTING

- Create a dynamic route by making a file:
  pages/products/[id].js
- Add the following code:

```
import { useRouter } from 'next/router';
export default function Product() {
  const router = useRouter();
  return <h1>Product ID: {router.query.id}</h1>;
}
```

- Run the app and visit:
- **http://localhost:3000/products/1**
- **http://localhost:3000/products/2**

DEMO: API ROUTES

- Create an API route in **pages/api/hello.js**
- Add the following code:

```
export default function handler(req, res) {
  res.status(200).json({ message: 'Hello, Next.js!' });
}
```

- Start the server (**npm run dev**)
- Visit the API endpoint:
- **http://localhost:3000/api/hello**

FETCHING DATA WITH GETSTATICPROPS (SSG)

- Use **getStaticProps** for static site generation
- Example:

```
export async function getStaticProps() {
  const res = await fetch('https://jsonplaceholder.typicode.com/posts');
  const posts = await res.json();
  return { props: { posts } };
}
```

- **Benefits of SSG:**
- ✅ Faster page load
- ✅ Content generated at build time

FETCHING DATA WITH GETSERVERSIDEPROPS (SSR)

- Use **getServerSideProps** to fetch data on each request
- Example:

```
export async function getServerSideProps() {
  const res = await fetch('https://jsonplaceholder.typicode.com/posts');
  const posts = await res.json();
  return { props: { posts } };
}
```

- **Use Cases:**
- Real-time user-specific data
- Frequent database updates

INCREMENTAL STATIC REGENERATION (ISR)

- ISR allows you to update static content without rebuilding the entire site
- Example:

```
export async function getStaticProps() {
  const res = await fetch('https://jsonplaceholder.typicode.com/posts');
  const posts = await res.json();
  return { props: { posts }, revalidate: 10 };
}
```

- **How it works?**
- Rebuilds the page **every 10 seconds**
- Old pages are cached until revalidation

NEXT.JS MIDDLEWARE

- Middleware allows handling requests before rendering a page
- Example:

```
import { NextResponse } from 'next/server';
export function middleware(req) {
  if (!req.cookies.auth) {
    return NextResponse.redirect('/login');
  }
}
```

- **Use Cases:**
- Authentication
- Redirects
- Custom request logging

STYLING IN NEXT.JS

- Methods of styling in Next.js:
- ✅ Global CSS (styles/globals.css)
- ✅ CSS Modules (Component.module.css)
- ✅ Styled-components
- ✅ Tailwind CSS
- Example using CSS Modules:

```
.button { background: blue; color: white; padding: 10px; }
```

```
import styles from '../styles/Button.module.css';
export default function Button() {
  return <button className={styles.button}>Click Me</button>;
}
```

USING TYPESCRIPT IN NEXT.JS

Enable TypeScript:

```
touch tsconfig.json && npm install typescript @types/react
```

Rename files to .tsx
Example Component:

```
type Props = { name: string };
const Welcome: React.FC<Props> = ({ name }) => <h1>Welcome, {name}</h1>;
export default Welcome;
```

Benefits of TypeScript:
✅ Type safety
✅ Better developer experience

THANK YOU

FOR YOUR ATTENTION